Customer Shopping Behavior Analysis - Retail Business

1. **Project Overview**

   Purpose:

   - Analyze customer shopping behavior to uncover trends, improve engagement, and optimize marketing and sales strategies.

   Business Problem:

   - How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

2. **Dataset Summary**

   - Records: 3,900 customers
   - Columns: 18
   - Key Features:
     - Customer demographics (Age, Gender, Location, Subscription Status)
     - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
     - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
     - Missing Data: 37 values in *Review Rating* column were replaced with the median of each product category

3. **Exploratory Data Analysis (Python)**

   We began with data preparation and cleaning in Python:

   - **Data Loading:** Imported the dataset using pandas.

   - **Initial Exploration:** Used df.info() to check structure and df.describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the ''Review Rating'' column, using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

    - Created **age_group** column by binning customer ages.
    - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.

- **Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.**

4. **Data Analysis using SQL (Business Transactions)**
   Performed structured analysis in PostgreSQL to answer key business questions:

A. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

B. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79

C. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Blouse	3.68
Sweater	3.76
Jeans	3.65
Sandals	3.84
Sneakers	3.76

D. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	Average spend
Express	60.48
Free Shipping	60.41
Next Day Air	58.63
Standard	58.46
2-Day Shipping	60.73
Store Pickup	59.89

E. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

F. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00000
Sneakers	49.65517
Coat	49.06832
Sweater	48.17073
Pants	47.36842

G. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	No. of Customers	segment_percentage
Loyal	3116	79.90
Returning	701	17.97
New	83	2.13

H. **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

I. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

J. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

6. Business Recommendations

I. Diversify Revenue Across Categories

Revenue is heavily concentrated in Clothing, while Footwear and Outerwear underperform.

Recommendation:

- Introduce targeted promotions and discounts for low-performing categories
- Bundle underperforming items with best-selling products (e.g., shirts + sneakers)
- Run seasonal campaigns for outerwear

Goal: Reduce dependency on a single category and increase overall revenue balance.

II. Leverage Top-Selling Products

Products like shirts, scarves, and sunglasses are driving the most revenue.

Recommendation:

- Increase inventory and visibility of top-performing products
- Use them in cross-selling strategies (e.g., "Customers also bought…")
- Feature them prominently in marketing campaigns

Goal: Maximize revenue from proven high-demand items.

III. Improve Underperforming Products

Footwear (especially sneakers) shows the lowest sales among top products.

Recommendation:

- Review pricing strategy and competitor positioning
- Improve product descriptions, images, and promotions
- Collect customer feedback to identify issues

Goal: Increase sales performance in weaker product lines.

IV. Target High-Value Customer Segments

Young adults generate the highest revenue among age groups.

Recommendation:

- Focus marketing efforts on young adults (social media, influencer marketing)
- Personalize offers based on customer preferences and behavior
- Maintain engagement strategies for other age groups to sustain balance

Goal: Maximize returns from the most valuable customer segment.

V. Enhance Customer Satisfaction

Average rating of 3.75 suggests room for improvement.

Recommendation:

- Improve product quality and consistency
- Optimize delivery and shipping experience
- Strengthen customer support and feedback systems

Goal: Increase customer retention and long-term value.

VI. Optimize Based on Rating–Revenue Relationship

Revenue is strongest between 3.5 - 4.5 ratings, not necessarily perfect scores.

Recommendation:

- Focus on improving poorly rated products (<3.0)
- Maintain quality consistency rather than chasing only 5-star ratings
- Identify high-revenue products with lower ratings and improve them

Goal: Protect revenue while improving overall customer experience.

Overall, the business should focus on strengthening high-performing categories, improving weaker segments, and enhancing customer experience to drive sustainable revenue growth and customer retention.